

FWX AVIATION

Helicopter Flight Dynamics

Live flight dynamics for every helicopter
in Microsoft Flight Simulator 2024

CONTENTS

What you will find in this manual

- 01 The idea behind the product
- 02 Getting started
- 03 The panel at a glance
- 04 Profiles and the master switch button
- 05 Main rotor and tail rotor
- 06 Effect strength and response
- 07 Hover
- 08 Flight in motion
- 09 Factory settings and how to return to them
- 10 If the effect feels too strong or too weak
- 11 Quick reference

Everything you feel is calculated live

Nothing in this product is a recording, a script or a random generator. Every movement is worked out moment by moment from what the simulator reports about your helicopter and the air around it. Turn the weather off and the aircraft settles down. Fly into a gusty afternoon and it comes alive. That connection is the whole point of the product.

A helicopter that answers the air

A real helicopter is never completely still. It reacts to the air it flies through, to its own rotor and to the hand on the controls. This product adds exactly that layer to Microsoft Flight Simulator 2024, for every helicopter you own and not just one type.

The module runs quietly in the background and works together with the flight model of the aircraft rather than replacing it. You keep the handling you know; what you gain is the sense that the machine is standing in moving air instead of on rails.

Which live values are used

The module continuously reads the simulator's own values while you fly. Among them are:

- **Airflow around the aircraft:** wind direction and strength, crosswind, the airflow from ahead, and how quickly they change.
- **Gusts and weather:** what counts is the change in the air, not a fixed pattern. Calm weather produces a calm aircraft.
- **Rotor speed:** read from the aircraft itself, so a spooling rotor behaves differently from one at full speed.
- **Your control inputs:** cyclic, collective and pedals, including how fast you move them.
- **The state of flight:** airspeed, attitude, vertical movement and height above the ground.
- **The aircraft itself:** the mass and inertia the simulator reports for the helicopter you have loaded.

Because all of this is read live and used live, the same helicopter feels different on a still morning than it does in a squall, and a light two-seater never feels like a heavy transport machine.

What stays yours

The module never takes control away from you. It adds motion; it does not fly the aircraft. Every slider in this manual can be turned down to nothing, and the master switch turns the whole thing off instantly.

Your first five minutes

Install the package, start the simulator, load a helicopter and open the FWX Control Center from the toolbar. That is all that is needed, because the module starts with the simulator by itself.

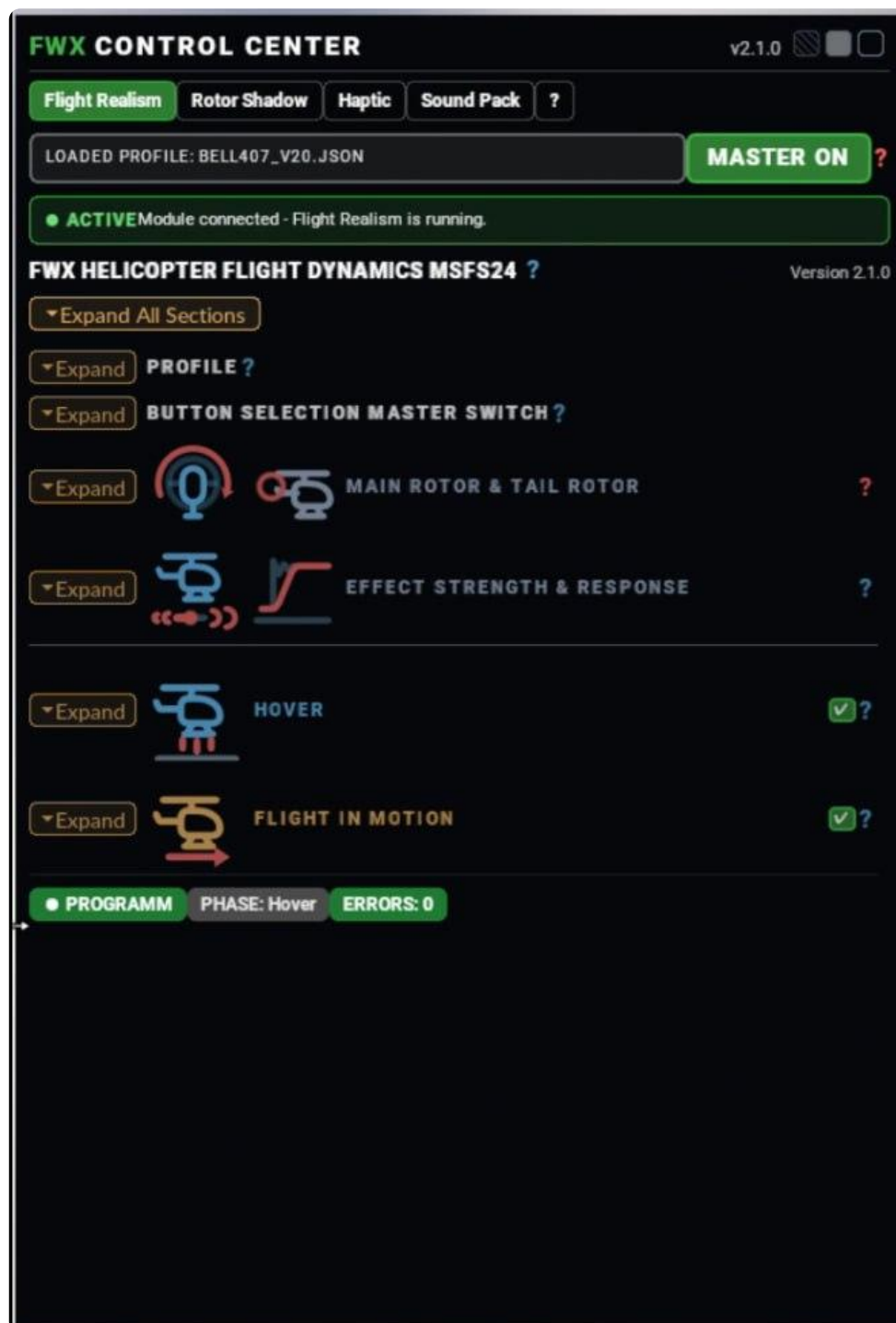
Step	What to do
1. Open the panel	Click the FWX icon in the simulator toolbar. The Flight Realism tab is the one described in this manual.
2. Check the status	A green ACTIVE bar with “Module connected” means everything is running. If it is grey, the module is installed but not connected. Check that it is allowed to start with the simulator.
3. Switch it on	Press MASTER ON . This is the main switch for the whole product.
4. Fly	Lift into a hover and hold it. You should feel the aircraft working gently against you instead of sitting still.
5. Adjust	Open a section and move one slider at a time. Changes take effect immediately, while you fly.

One helicopter, one setting

Every setting you make belongs to the helicopter type you are flying. Set up your light training machine the way you like it, then load a heavy transport type. It keeps its own values. Nothing you tune on one aircraft changes another.

The panel at a glance

The panel is built as a short list of sections. Each one can be folded open or closed, so you only see what you are working on.



The Flight Realism tab with all sections closed

Element

What it does

Loaded profile	Shows which set of values is currently in use.
MASTER ON / OFF	The main switch. Off means the module produces nothing at all.
Status bar	Green means the module is connected and running.
Expand all sections	Opens or closes every section in one go.
Question marks	Hold the mouse over one and a short explanation appears.
Check boxes	Switch a single item off. It then falls back to the simulator's own original behaviour, and your value is kept.
Reset symbol	Returns that one slider to its factory value.

Profiles and the master switch button

A profile holds every setting for one helicopter type. You can keep several of them and swap between them while you fly.



Saved profiles and the button assignment for the master switch

Button	What it does
LOAD PROFILE	Loads a saved profile into the panel and into the module. Your sliders jump to the values stored in it.
SAVE PROFILE	Saves the current sliders to the helicopter you are flying. If a profile from another helicopter is loaded, you are asked what you want: save into the profile of the helicopter you are sitting in, or create a new profile. The profile you borrowed is never overwritten.
SAVE AS...	Stores the current sliders under a name of your own, so you can keep a calm setting and a lively one side by side.
FACTORY SETTINGS	Puts every slider back to the values the product ships with. It takes effect at once; it becomes permanent when you save.

You do not start from zero

Every helicopter type that Asobo ships with Microsoft Flight Simulator 2024 comes with a tested default profile. When you load one of those helicopters for the first time, proven settings are already in place. From there you only refine them to your own taste.

Assigning a button to the master switch

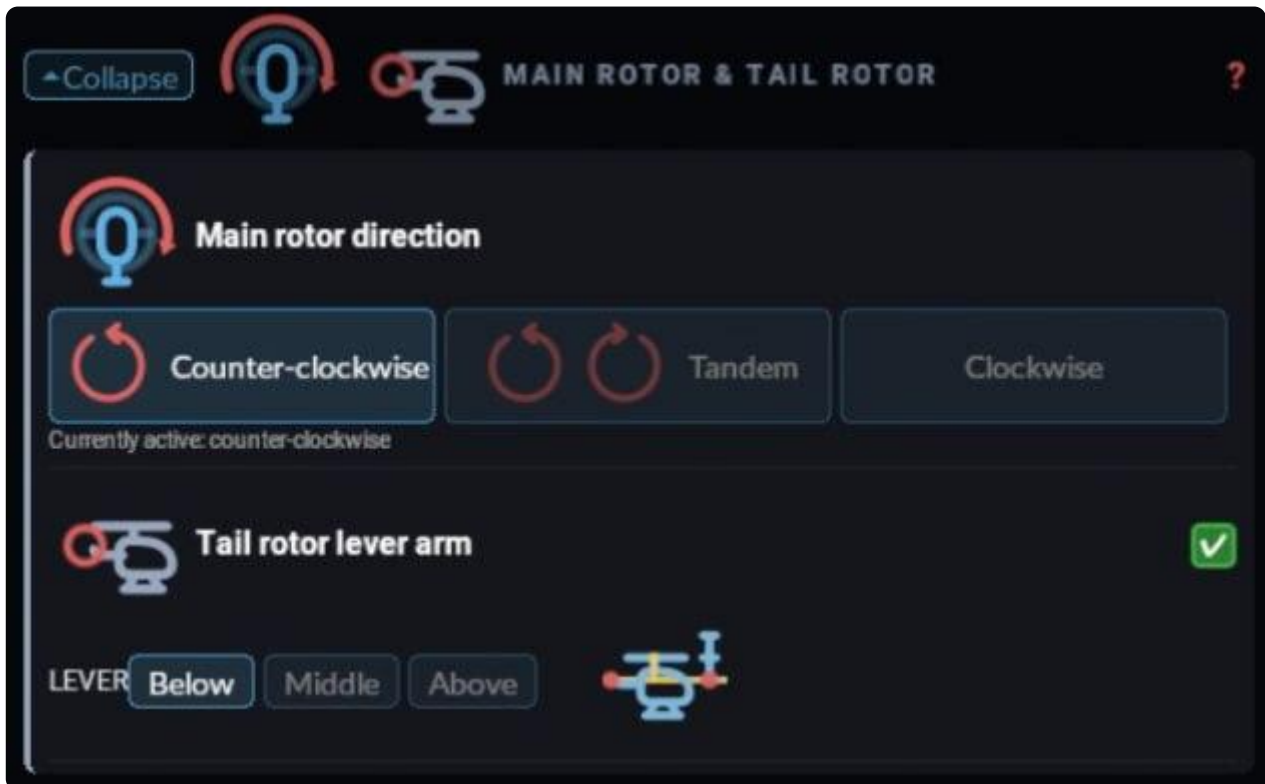
Press **Assign button** and then press the key, mouse button or joystick button you would like to use. From then on, that button switches the whole module on and off while you fly, without opening the panel. **Clear** removes the assignment again.

Unsaved changes

While a setting differs from the last saved state, the panel says so. Save it if you want to keep it; load the profile again if you would rather discard it.

Main rotor and tail rotor

These two settings describe how the helicopter you are flying is actually built. They matter more than any slider, because everything that follows depends on them.



Main rotor direction and tail rotor lever arm

Main rotor direction

Choose the direction in which the main rotor of this helicopter turns, seen from above: **counter-clockwise**, **clockwise**, or **Tandem** for machines with two main rotors turning opposite ways, such as the Chinook or the Kamov types. The line underneath always shows which direction is currently in use.

Please check this first

Does the main rotor of this helicopter really turn the way you picked here? If the direction is wrong, every effect that depends on it works the other way round. With a tandem there is no torque yaw and no tail rotor, so the lever arm below is set to the middle and locked.

Tail rotor lever arm

This describes where the tail rotor sits relative to the centre of gravity. **Above** means the tail rotor sits above the centre of gravity and rolls the helicopter in the direction of its thrust; **Below** is the opposite case; **Middle** means the lever arm has no effect. Look at your aircraft and pick what you see. If you switch the lever arm off entirely, airflow, gusts and ground-effect rocking still work.

Effect strength and response

Three dials for the character of the whole aircraft: how strong it feels, how softly it arrives, and how quickly it builds up.



Overall strength, output smoothing and build-up speed

Control	What it does
Overall Strength of Effect	One dial for the whole helicopter. It scales every value of this helicopter type together. 100 % is normal, higher is stronger, which suits light and agile types. The block strength sliders further down scale one block only.
Output smoothing	Sets how softly the effects arrive. Lower feels sharper and faster, higher feels softer. Light helicopter types react more quickly, so a shorter smoothing often suits them better.
Effect build-up speed	Sets how quickly an effect may build up. Raise it and short, sharp impulses come through more completely; lower it and everything arrives more gently.

Each of these has a check box. When it is cleared, that part of the module is switched off and the simulator's own original behaviour applies again, exactly as Asobo built it. Your value is kept safely and comes back when you switch it on again.

Hover

This block shapes how the helicopter behaves while it is hovering or moving very slowly. The change to the block below is a seamless transition, so the two never fight each other.



The complete hover section: block strength, gusts and the three axes

The controls in the hover block

Control	What it does
Block strength	Sets how clearly the helicopter feels alive in this block. It scales all three axes by the same ratio. At 0, nothing in this block moves any more. This is purely a matter of taste.
Gusts	Short pushes from the air, driven by the change in the wind rather than by your stick. In still weather it produces nothing at all.
Pitch	The nose tips up and down.
Yaw	The nose swings left and right.
Roll	The aircraft leans left and right.
Ground effect	Sets how clearly the helicopter notices the nearness of the ground, which is the rocking and working you feel when you are hovering close above it. Turn it down for a calmer machine near the ground, turn it up if you want to feel the surface more. The value belongs to the profile of this helicopter type and is saved with every other slider.

The **Ground effect** slider sits in the hover section together with the controls listed above. It was hidden at the time the screenshots in this manual were taken, so it does not appear in the picture on the previous page.

Gusts acts on all three axes, that is on pitch, yaw and roll. **Ground effect** acts on pitch and roll.

The two sliders on every axis

Slider	What it does
Airflow	The part of the motion that comes from the air around the aircraft: wind, crosswind and how it changes. Raise it and weather makes itself felt more clearly.
Mechanical Dynamics	The part that comes from the machine itself, meaning the rotor and the way the airframe answers it. Raise it and the helicopter feels more mechanical and alive even in calm air.

A good way to tune it

Work on one axis at a time. Set both sliders of that axis to zero, then raise Airflow until the weather feels right, and only afterwards raise Mechanical Dynamics until the machine feels right. Doing both at once makes it hard to tell which one you are hearing.

Flight in motion

The same three axes again, but for flight in any direction once the helicopter is moving. The transition from the hover block is seamless.



The complete flight-in-motion section, including speed gain

What is different here

Control	What it does
Block strength	As in the hover block: it scales all three axes of this block together.
Speed gain	The faster the helicopter flies, the more strongly the air makes itself felt. In the hover this slider does nothing at all. Leave it at 0 and everything stays as it is; raise it in small steps until fast flight feels right to you.
Gusts	As in the hover block: short pushes driven by the change in the air.
Pitch, Yaw, Roll	The same three axes, each with its own Airflow and Mechanical Dynamics slider.

Because the two blocks are separate, you can have a calm, precise hover together with a lively cruise, or the reverse. Many pilots end up with a little more Airflow in motion and a little more Mechanical Dynamics in the hover.

Returning to a known state

Nothing you do can be lost for good. There are three ways back, from a single slider to the whole aircraft.

What you want	How to do it
One slider back	Press the small reset symbol next to that slider. Only that value changes.
One item off, value kept	Clear its check box. That part falls back to the simulator's own original behaviour, and your value stays where it is until you switch it on again.
The whole helicopter back	Press FACTORY SETTINGS in the profile section. Every slider returns to the shipped values.
Everything off at once	Press MASTER OFF . The module produces nothing until you switch it on again.

Immediate, but not permanent

A reset takes effect while you fly, but it is only written to disk when you save the profile. If you change your mind, load the profile again and your previous values come back.

The product ships with a prepared set of values for every helicopter type it knows. If you load a type it has not seen before, it starts from a sensible neutral setting, which you are then free to shape.

If it feels too strong or too weak

The aircraft moves too much

- Lower **Overall Strength of Effect** first. One dial, whole aircraft.
- If it is only one flight regime, lower the **block strength** of that block instead, either hover or flight in motion.
- If it feels nervous rather than strong, raise **Output smoothing** and lower **Effect build-up speed**. The same motion then arrives more gently.
- If it only happens in rough weather, lower the **Gusts** slider of that block.
- If it only happens at speed, lower **Speed gain**.

The aircraft feels lifeless

- Check first that **MASTER** is on and the status bar is green.
- Check the weather. In perfectly still air there is very little for the airflow sliders to work with. This is correct behaviour, not a fault.
- Raise **Mechanical Dynamics** on the axis that feels dead. That part does not depend on the weather.
- Raise **Overall Strength of Effect** in small steps. Light and agile types often want more than the default.
- Make sure the check boxes of the sliders you are moving are actually ticked.

Something feels reversed

- Check the **main rotor direction**. A wrong direction reverses every effect that depends on it.
- Check the **tail rotor lever arm**. Above and below produce opposite roll behaviour.

Change one thing at a time

Move a single slider, fly a short circuit, then decide. If you change two things at once, you will not know afterwards which one did the work.

Quick reference

Control	In one sentence
MASTER	The main switch for the entire product.
Main rotor direction	Which way the main rotor turns, seen from above.
Tail rotor lever arm	Where the tail rotor sits relative to the centre of gravity.
Overall Strength of Effect	One dial for the whole helicopter.
Output smoothing	How softly the effects arrive.
Effect build-up speed	How quickly an effect reaches full strength.
Block strength	Scales all three axes of one block together.
Gusts	Short pushes from changing air.
Speed gain	More effect the faster you fly. In the hover it does nothing.
Airflow	The part of the motion that comes from the air.
Mechanical Dynamics	The part that comes from the machine itself.
Reset symbol	Returns one slider to its factory value.
FACTORY SETTINGS	Returns the whole helicopter to the shipped values.

A short checklist

Set the rotor direction and the lever arm to match the helicopter. Decide how strong the whole aircraft should feel. Then set up the hover and the flight in motion separately. Everything else is a matter of taste, and every change can be undone.

Questions and suggestions

If something is unclear, if you run into a problem or if you have an idea for us, we are glad to hear from you. You will find how to reach us in the panel itself: open the question mark tab in the FWX Control Center, where our contact details are listed.

